Shu Yang

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Research Interests & Personal Background

Shu Yang is an undergraduate student in the Excellence Class of Computer Science at Nankai University. During high school, he won a bronze medal at the National Olympiad in Informatics (NOI 2022). After entering university, he continued to participate in ACM-ICPC/CCPC programming contests, winning multiple medals, and possesses solid skills in algorithm design and implementation. His long-term experience in programming contests has sparked a strong interest in the theoretical foundations of computational complexity and the intrinsic laws of algorithm design. He aspires to pursue foundational research in algorithm design in the future.

During his undergraduate studies, he has systematically taken theoretical courses such as *Computational Theory*, *Lattice-based Post-Quantum Cryptography*, *Algorithm Design and Analysis*, and *Randomized Algorithms*, as well as engineering practice courses like *Operating Systems*, *Compilers*, *Database Systems*, and *Computer Graphics*. He is capable of analyzing and solving complex problems, and also possesses engineering coding ability. Additionally, as an undergraduate research assistant, under the supervision of Associate Professor Shenwei Huang and Dr. Yidong Zhou, he participated in research on the solvability of graph coloring problems, gaining experience in literature reading, algorithmic reduction, and complexity analysis.

Education

Nankai University, College of Computer Science, Computer Science and Technology (Excellence Class) *B.S. (expected)* 2023.09 – Present

- GPA: 3.48/4.0, Rank: Top 38%

Affiliated High School of Shanxi University, Experimental Class 2020.09 – 2023.06

- Participated in informatics competitions during high school, ranked third in the province and selected for the Shanxi provincial team, won a bronze medal at NOI, laying the foundation for subsequent university studies.

Research Experience

Complexity of 3-Coloring on a Special Graph Class (Supervisors: Associate Professor Shenwei Huang, Dr. Yidong Zhou)

- **Goal**: Determine whether the 3-coloring problem on a class of graphs with specific structural constraints is in P.
- **Main Work**:
  - Reviewed and summarized core papers on the complexity boundary of graph coloring from the past decade, wrote literature review notes.
  - Attempted to design polynomial-time algorithms: used graph decomposition techniques to reduce the target graph class, initially proposed a 3-coloring algorithm framework.
  - Explored reduction proofs: constructed polynomial reductions from known NP-complete problems (e.g., 3-SAT or Not-All-Equal 3-SAT) to this graph class.
- **Academic Communication**: Attended weekly group seminars, reported research progress, and discussed verification methods for correctness of reductions with supervisors.

Course Projects

Implementation of a Mini Compiler for SysY

- Implemented a front-end compiler for the SysY language (a C subset) using C++ and Flex/Bison: lexical analysis, syntax analysis, semantic analysis, code optimization, and intermediate code generation.
- Designed symbol tables and type checking modules, implemented intermediate representation for basic control flow and function calls.

Database System Based on the CMU BUSTUB Framework

- Implemented in-memory/disk dual storage model using C++, supporting B+ tree indexes, a SQL subset (SELECT, INSERT, DELETE), and ACID transaction properties (simple log recovery).
- Buffer LRU replacement policy; performance comparison test reports for single-table scans showed an approximately 40x speedup for indexed queries.

Real-time Ray Tracing Renderer

- Implemented ray tracing in C++, supporting multiple light sources, shadows, specular reflection, and refraction.
- Used KD-Tree to accelerate intersection tests, greatly improving rendering speed.

All projects above are uploaded to my GitHub repository.

Competition Experience

In the ACM three-person team, I am mainly responsible for dynamic programming, graph theory, combinatorial counting, and some data structure design. I excel at problems requiring algorithmic thinking.

ACM-ICPC International Collegiate Programming Contest 2025

- Shenyang Regional: Silver medal (team rank 2).
- Hong Kong Regional: Silver medal (team rank 3).
- EC Final: Bronze medal.

CCPC China Collegiate Programming Contest 2025

- Zhengzhou Site: Silver medal; CCPC Final: Bronze medal.

National Olympiad in Informatics (NOI 2022) 2022

- Bronze medal.

Technical Skills

- **Programming Languages**: Proficient in C/C++, Python.
- **Tools & Environments**: Linux (Ubuntu), Git, VS Code, LaTeX.
- **Theoretical Tools**: Algorithm complexity analysis, reduction proofs, combinatorics, basics of probabilistic methods.

English Proficiency

IELTS Overall Score 7.0 April 2025

- Listening 7.5, Reading 8, Writing 6.5, Speaking 6.5.
- Capable of reading English academic papers fluently and writing technical documents.

Honors and Awards

- Bronze Medal, National Olympiad in Informatics (NOI 2022)
- Multiple Medals, International Collegiate Programming Contest (ICPC) / China Collegiate Programming Contest (CCPC)